

The Mars Sample Return Ascent Vehicle

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Abstract

The multi-purpose Mars Sample Return mission is intended to retrieve the cores that Perseverance will gather during the following several years. The project, which is one of the most ambitious attempts in the history of spaceflight and involves several spacecraft, multiple launches, and dozens of government entities, is now in the concept design and technological development phase.

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1 Constants and Useful Parameters

Only SI units are used:

$$G_{\text{grav}} = 6.67408 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$$

$$c_{\text{Light}} = 2.99792458 \times 10^8 \text{ m/s}$$

2 Introduction

According to a plan now suspended and under budget review, a Sample Retrieval Lander will be sent to Mars together with a NASA-led Mars rocket and two miniature Mars helicopters. In Jezero Crater, the lander will touchdown near the location of the Perseverance landing. The Perseverance rover will be the main vehicle for delivering samples to the Sample Retrieval Lander, handing off a collection of sample tubes carried on board, having gathered and stored a wide range of materials. The tubes will be moved into a sample container on the lander's rocket by a robotic arm supplied by the ESA. A supplementary capability will be provided by two helicopters, capable of picking up extra samples that Perseverance had hidden on the surface and return them to the lander for loading aboard the Mars rocket. The container of sample tubes will be carried into orbit by the Mars Ascent Vehicle, the first rocket to ever launch from the surface of Mars. The NASA-supplied Capture, Containment, and Return System and Earth Entry System would be mounted aboard the ESA Earth Return Orbiter. The samples would be captured and contained by the orbiter's return system, which would then deposit them in the Earth Entry System. The entrance vehicle would subsequently be transported back to Earth's vicinity by the Earth Return Orbiter, where it would detach and land safely [?].

3 Mars Ascent Vehicle

3.1 Components

The main components of the Mars Ascent Vehicle (MAV) include:

- **Propulsion System:**
 - Fuel: Monomethyl Hydrazine (MMH)

- Oxidizer: a mixture of Nitrogen Tetroxide and Nitrogen Monoxide (MON-25)
- **Structural Components:**
 - Payload compartment
 - Thrust structure and engine mount
 - Aerodynamic fairings
- **Avionics:**
 - Guidance and navigation systems
 - Communication modules
- **Life Support (if manned):**
 - Oxygen supply
 - Temperature and pressure control

3.2 Duties

The primary duties of the MAV are:

- **Lift-off from Mars surface:** Provide sufficient thrust to overcome Mars gravity

$$g_{\text{Mars}} = 3.728 \text{ m/s}^2$$

- **Achieve Mars orbit insertion:** Reach desired orbital velocity

$$v_{\text{orbit}}$$

- **Payload delivery:** Safely transport cargo or samples to orbit or rendezvous with a Mars Orbiter.
- **Ensure system reliability:** Maintain structural and operational integrity during ascent.

4 Sample Retrieval Lander

4.1 Components

The main components of the Sample Retrieval Lander (SRL) include:

- **Landing System:**
 - Retrorockets for soft landing
 - Landing legs and shock absorbers
- **Sample Handling System:**
 - Robotic arm for collecting sample tubes
 - Sample storage compartment

- Environmental control system to maintain sample integrity
- **Avionics:**
 - Guidance, navigation, and control (GNC)
 - Communication systems to relay data to orbiters and Earth

4.2 Duties

The SRL is responsible for:

- **Receiving samples:** Safely accept pre-collected sample tubes from Perseverance rover [?].
- **Sample containment:** Secure samples in temperature- and pressure-controlled compartments.
- **Prepare for ascent:** Transfer the samples to the Mars Ascent Vehicle (MAV) for return to orbit.
- **Maintain system integrity:** Ensure all mechanisms function reliably during landing, transfer, and ascent operations.

5 Mars Ascent Vehicle Solid Propellant Engines

Two solid fuel engines are used by the Mars Ascent Vehicle. The fuel for it will be TP-H-3062. The oxidizer-fuel ratio is 1.27 [?].

6 Staging

The Mars Ascent Vehicle employs a two-stage configuration to optimize mass efficiency and ascent performance under Martian gravity.

6.1 First Stage

- Provides the initial thrust required for lift-off from the Martian surface.
- Overcomes Mars gravity and atmospheric drag.
- Operates for a short, high-thrust burn period.
- Is jettisoned after propellant depletion to reduce dead mass.

6.2 Second Stage

- Ignites immediately after first-stage separation.
- Responsible for achieving orbital velocity and orbit insertion.
- Carries the payload (scientific samples) into Mars orbit.
- Optimized for higher efficiency and precise trajectory control.

This staging strategy minimizes total vehicle mass while maximizing ascent efficiency and mission reliability [?].

7 Aerodynamic Stress

During ascent through the Martian atmosphere, the Mars Ascent Vehicle (MAV) is subjected to aerodynamic stress due to the relative motion of air and vehicle surfaces. The primary aerodynamic forces are drag and dynamic pressure, which must be carefully considered to prevent structural failure.

7.1 Dynamic Pressure

The dynamic pressure q is defined as:

$$q = \frac{1}{2} \rho v^2$$

where:

- ρ = atmospheric density on Mars [kg/m³]
- v = velocity of the vehicle relative to the atmosphere [m/s]

7.2 Drag Force

The drag force F_D acting on the vehicle is:

$$F_D = C_D q A_{\text{ref}}$$

where:

- C_D = drag coefficient (depends on vehicle shape)
- A_{ref} = reference area of the MAV [m²]

7.3 Structural Stress

The aerodynamic stress experienced by the MAV can be approximated as:

$$\sigma = \frac{F_D}{A_{\text{struct}}}$$

where A_{struct} is the cross-sectional area of the structural components experiencing the load.

8 Analysis and Results

8.1 Locating and Drawing the Main Axis of the Orbit and the Argument of the Periapsis

The main (major) axis of an orbit is defined by the line connecting the periapsis and apoapsis of the trajectory. Its orientation in inertial space is determined by the argument of periapsis, ω , measured from a reference direction. For a planar two-body orbit, the specific angular momentum is:

$$\mathbf{l} = \mathbf{r} \times \mathbf{v}$$

The eccentricity vector is given by:

$$\mathbf{e} = \frac{1}{k} (\mathbf{v} \times \mathbf{l})$$

The direction of the eccentricity vector points toward periapsis. Therefore, the argument of periapsis is obtained from:

$$\omega = \tan^{-1} \left(\frac{e_y}{e_x} \right)$$

The code for locating the argument of periapsis and plotting the major axis is as follows:

```
(* Position and velocity at orbit injection *)
rVec = {rSolSC[tfin] Cos[ThetaAngSolSC[tfin]], rSolSC[tfin] Sin[ThetaAngSolSC[tfin]], 0}
vVec = {rSCdotSol[tfin] Cos[ThetaAngSolSC[tfin]] - vtSolSC[tfin] Sin[ThetaAngSolSC[tfin]],
        rSCdotSol[tfin] Sin[ThetaAngSolSC[tfin]] + vtSolSC[tfin] Cos[ThetaAngSolSC[tfin]], 0}

(* Specific angular momentum *)
lVec = Cross[rVec, vVec];

(* Eccentricity vector *)
eVec = (Cross[vVec, lVec]/kMars) - (rVec/Norm[rVec]);

(* Argument of periapsis *)
omegaAOP = ArcTan[eVec[[1]], eVec[[2]]] * 180/Pi;
```

The major axis of the orbit aligns with the eccentricity vector. The argument of periapsis corresponds to the angle between this axis and the inertial reference direction. The plotted line visually confirms the analytical value of ω .

8.2 Dependence of the Periapsis Altitude on Initial Thrust Angle, ϕ_0

The periapsis altitude is plotted as a function of the initial thrust angle, ϕ_0 , to understand its effect on the orbit. For sufficiently low thrust angles, the periapsis radius becomes smaller than the Mars radius, indicating surface impact during the first orbit. The code for calculating and plotting this dependence is as follows:

```
(* Effective burn time for single stage *)
tfin = (MassRocket0*(1 - PayloadFraction - DryVehicleFraction))/mdot;

(* Thrust angle function *)
ThrustPhi1[t_, phi0_] := ArcTan[(1 - t/tfin) Tan[phi0 Degree]];

(* Function to compute periapsis altitude *)
periapsisAltitude[phi_?NumericQ] := Module[
  {sol, rFunc, thetaFunc, rdot, vt, energy, l, e, a, rp},
  sol = NDSolve[{rSC'[t] == rSC[t] (ThetaAngSC'[t]^2) - kMars/rSC[t]^2 + (mdot*vexh/(1
```

```

ThetaAngSC''[t] == -(2/rSC[t]) rSC'[t] ThetaAngSC'[t] +(1/rSC[t]) (md
rSC[0] == RadiusMars, rSC'[0] == 0, ThetaAngSC[0] == 0, ThetaAngSC'[0]
{rSC, ThetaAngSC}, {t, 0, tfin}, MaxSteps -> Infinity][[1]];

rFunc[t_] := Evaluate[rSC[t] /. sol];
thetaFunc[t_] := Evaluate[ThetaAngSC[t] /. sol];
rdot = rFunc'[tfin];
vt = rFunc[tfin] thetaFunc'[tfin];
energy = (rdot^2 + vt^2)/2 - kMars/rFunc[tfin];
l = rFunc[tfin] vt;
e = Sqrt[1 + (2 energy l^2)/kMars^2];
a = -kMars/(2 energy);
rp = a (1 - e);
rp - RadiusMars];

```

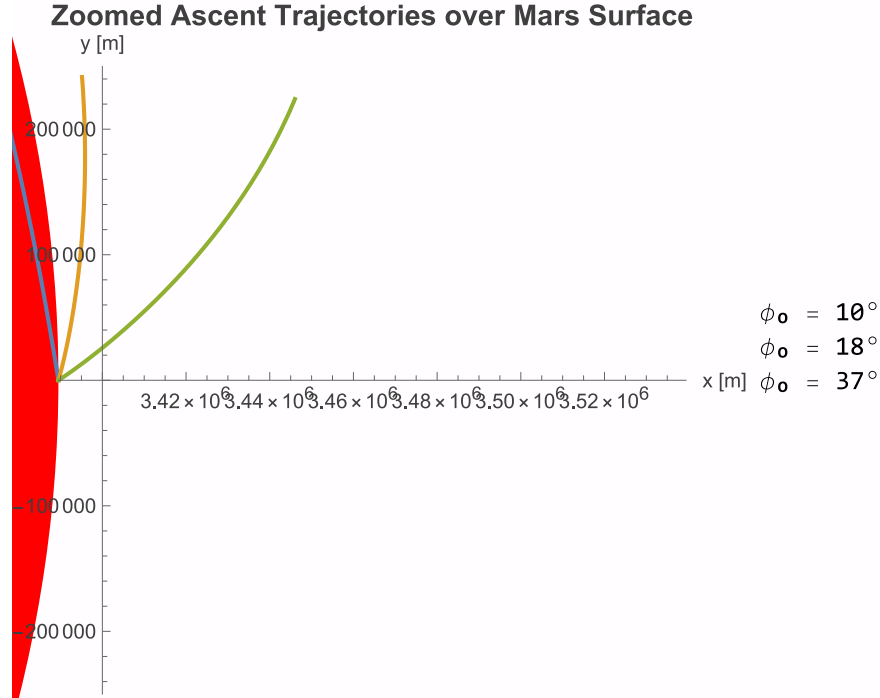


Figure 1: Periapsis altitude vs initial thrust angle ϕ_0 for sample launch configurations.

As the initial thrust angle increases, the periapsis altitude initially rises, reaching a maximum before decreasing. For low thrust angles, the periapsis radius falls below the Mars radius, suggesting impact on the surface.

8.3 Effect of Staging on Results

Staging increases performance by discarding inert mass, allowing the remaining propellant to accelerate a smaller vehicle mass. The effect of staging on the periapsis altitude and orbital energy is examined by comparing the results of single-stage and two-stage launches. The two-stage configuration provides enhanced performance for orbit insertion by minimizing dead weight.

```
(* Stage parameters *)
stageFraction = 0.5; (* each stage = 50\% of initial mass *)
tburnStage = (MassRocket0*stageFraction)/mdot;
tfin = 2*tburnStage; (* total burn time *)

(* Mass function depending on stage *)
MassStage[t_] := If[t <= tburnStage, stageFraction MassRocket0, MassRocket0*(1 - stageFraction)]

(* Function to compute periapsis altitude *)
periapsisAltitudeStage[phi_?NumericQ] := Module[{sol, rFunc, thetaFunc, rdot, vt, energy},
  sol = NDSolve[{rSC'[t] == rSC[t] (ThetaAngSC'[t]^2) - kMars/rSC[t]^2 + (FThrustLAS / (rSC[t]^2))
    ThetaAngSC'[t] == -(2/rSC[t]) rSC'[t] ThetaAngSC'[t] + (1/rSC[t]) (FThrustLAS / (rSC[t]^2))
    rSC[0] == RadiusMars, rSC'[0] == 0, ThetaAngSC[0] == 0, ThetaAngSC'[0] == phi
    {rSC, ThetaAngSC}, {t, 0, tfin}, MaxSteps -> Infinity][[1]];

  rFunc[t_] := Evaluate[rSC[t] /. sol];
  thetaFunc[t_] := Evaluate[ThetaAngSC[t] /. sol];
  rdot = rFunc'[tfin];
  vt = rFunc[tfin] thetaFunc'[tfin];
  energy = (rdot^2 + vt^2)/2 - kMars/rFunc[tfin];
  l = rFunc[tfin] vt;
  e = Sqrt[1 + (2 energy l^2)/kMars^2];
  a = -kMars/(2 energy);
  rp = a*(1 - e);
  rp - RadiusMars];
```

Atmospheric drag opposes the vehicle velocity and reduces the net acceleration during ascent. Although the Martian atmosphere is thin, drag losses can be significant during the early phase of flight when the vehicle velocity and atmospheric density are highest. The effect of drag is studied by comparing the periapsis altitude and orbital energy with and without aerodynamic drag.

```
(* Atmospheric model *)
CD = 0.2;
rRocket = 0.3;
Aeff = Pi rRocket^2;

kB = 1.380649*10^-23;
mu = 1.66053906892*10^-27;
H0 = (kB/mu) * 210 / (43.34 * gMars);

rhoAtm[r_?NumericQ] := 0.02 Exp[-(r - RadiusMars)/H0];
```

```
(* Drag force function *)
Fdrag[r_?NumericQ, rdot_?NumericQ, includeDrag_: True] := If[includeDrag, -0.5 CD Aeff
```

8.5 Staging with Unequal Stages and Atmospheric Drag

The final case considers unequal staging and atmospheric drag, which reduces the effectiveness of each stage. The vehicle performance is analyzed by comparing the periapsis altitude for different thrust configurations and stage parameters.

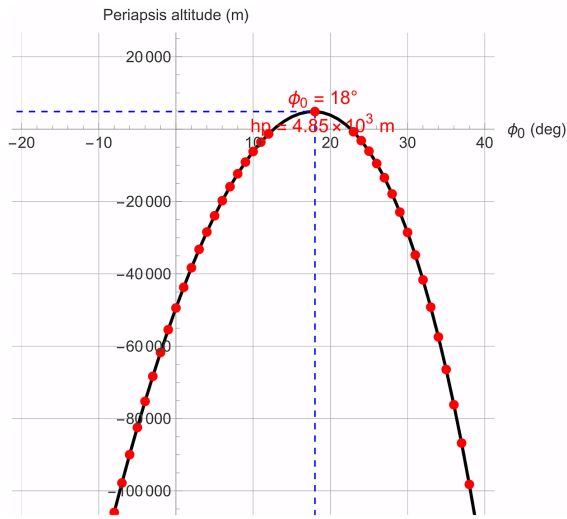
The code for this simulation is:

```
(* Stage parameters *)
stages = {
  <|"Mass0" -> 300., "FuelMass" -> 200., "ISP" -> 300, "Thrust" -> 8388|>,
  <|"Mass0" -> 150., "FuelMass" -> 100., "ISP" -> 300, "Thrust" ->5000|>};

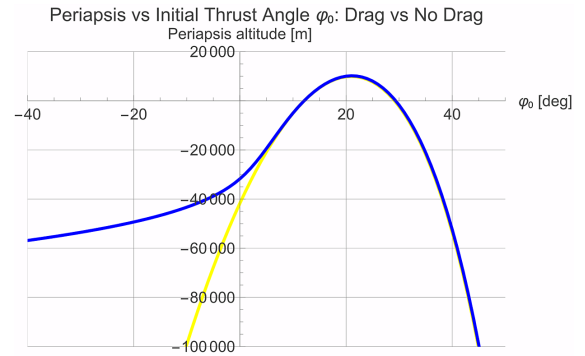
(* Simulate with different drag and thrust angle *)
periapsisAltitudeUnequalStage[phi_?NumericQ] := Module[{sol, rFunc, thetaFunc, rdot, vt},
  sol = NDSolve[{rSC'[t] == rSC[t] (ThetaAngSC'[t]^2) - kMars/rSC[t]^2 + Fdrag[rSC[t], rdot, phi],
    {rSC, ThetaAngSC}, {t, 0, tfin}, MaxSteps -> Infinity][[1]];
  rFunc[t_] := Evaluate[rSC[t] /. sol];
  thetaFunc[t_] := Evaluate[ThetaAngSC[t] /. sol];
  rdot = rFunc'[tfin];
  vt = rFunc[tfin] thetaFunc'[tfin];
  energy = (rdot^2 + vt^2)/2 - kMars/rFunc[tfin];
  l = rFunc[tfin] vt;
  e = Sqrt[1 + (2 energy l^2)/kMars^2];
  a = -kMars/(2 energy);
  rp = a*(1 - e);
  rp - RadiusMars];
```

8.6 Results and Graphs

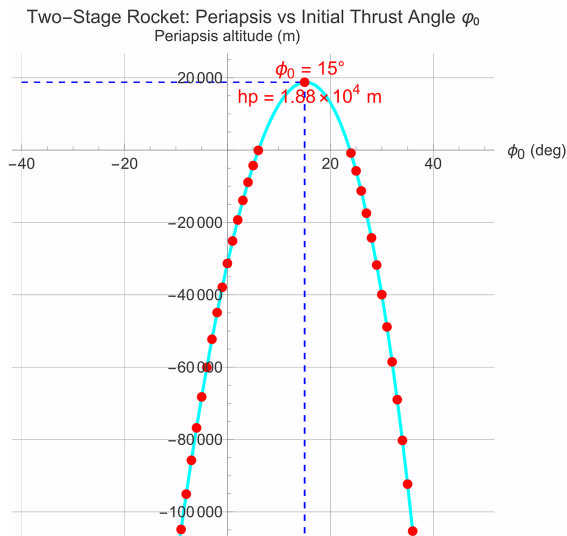
The results for periapsis altitude as a function of initial thrust angle, ϕ_0 , with varying staging configurations, drag models, and thrust angles, are plotted in the following figures.



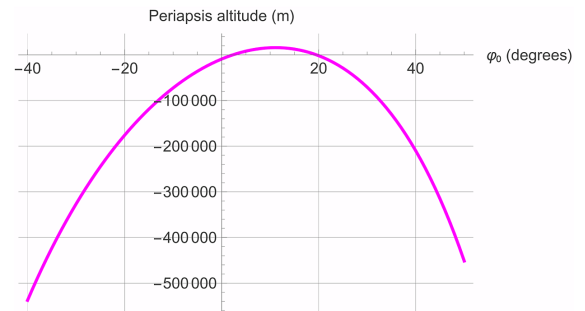
(a) Periapsis altitude vs initial thrust angle ϕ_0 .



(b) Effect of staging on periapsis altitude



(c) Effect of staging on periapsis altitude.



(d) Periapsis altitude with unequal stages and atmospheric drag.

Figure 2: Comparison of periapsis altitude under different configurations and conditions.

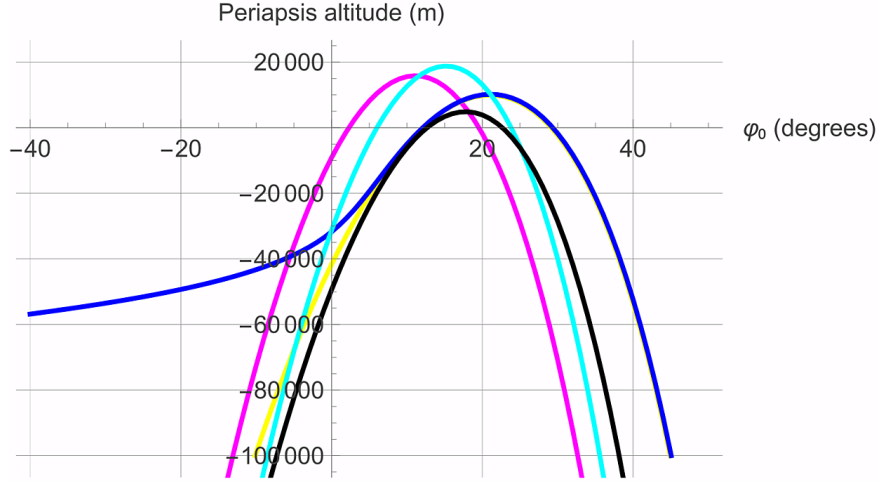


Figure 3: Periapsis altitude vs initial thrust angle ϕ_0 for various launch configurations.

9 Conclusions

From the analysis, the following conclusions are drawn:

- The periapsis altitude depends significantly on the initial thrust angle, ϕ_0 , with a higher thrust angle resulting in higher altitude.
- Staging improves the vehicle's performance by reducing the mass of the rocket during ascent, leading to a higher periapsis.
- Atmospheric drag plays a critical role in reducing the effective altitude by opposing the rocket's motion, especially in the early phase of flight.
- Unequal staging with drag reduces the periapsis altitude further, but a two-stage vehicle still outperforms a single-stage configuration.

10 References

References

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